

Teacher Pack: Primary Mathematics Diagnostic Pre-Test

Source reference: Singapore MOE *Primary Mathematics Syllabus* (2021 syllabus for P1-P6, updated October 2025). Topic choices prioritise P1-P4 syllabus content, with a small amount of P4 readiness.

B. Teacher Answer Key

Question	Answer	Skill Category	Approx. Level	What It Checks
1	6	Number Sense and Place Value	P1	Counts objects in a set accurately.
2	16	Number Sense and Place Value	P1	Continues a simple number sequence.
3	83	Number Sense and Place Value	P1	Compares two-digit numbers.
4	60	Number Sense and Place Value	P2	Understands place value in a 3-digit number.
5	406, 460, 640	Number Sense and Place Value	P2	Orders 3-digit numbers from smallest to greatest.
6	4,280	Number Sense and Place Value	P4 readiness	Rounds a 4-digit number to the nearest 10.
7	8	Addition and Subtraction	P1	Adds within 10.
8	8	Addition and Subtraction	P1	Subtracts within 20.
9	57	Addition and Subtraction	P1/P2	Adds a 2-digit number and tens mentally.
10	83	Addition and Subtraction	P2	Adds two 2-digit numbers with regrouping.
11	217	Addition and Subtraction	P2/P3	Subtracts 3-digit numbers accurately.
12	745	Addition and Subtraction	P3	Subtracts 4-digit numbers using algorithm.
13	6	Multiplication and Division Readiness	P1	Understands multiplication as equal groups.
14	20	Multiplication and Division Readiness	P2	Recalls multiplication facts.
15	6	Multiplication and Division Readiness	P2	Understands division as sharing.
16	56	Multiplication	P3	Recalls higher

		and Division Readiness		multiplication facts.
17	7 each, 1 left	Multiplication and Division Readiness	P3	Solves division with remainder.
18	1/4	Fractions	P2	Recognises a fraction as part of a whole.
19	1/2	Fractions	P2	Compares unit fractions.
20	1	Fractions	P3	Finds an equivalent fraction.
21	1/2	Fractions	P3	Simplifies a fraction.
22	1 3/4	Fractions	P4	Converts an improper fraction to a mixed number.
23	90c	Money and Time	P1/P2	Counts money in cents.
24	\$3.45	Money and Time	P2	Writes money in decimal notation.
25	\$0.50	Money and Time	P3	Subtracts money in decimal notation.
26	9:30 am	Money and Time	P1	Finds time 1 hour later.
27	45 minutes	Money and Time	P3	Finds duration using 24-hour time.
28	C or m	Measurement	P1/P2	Chooses an appropriate unit of length.
29	200	Measurement	P3	Converts metres to centimetres.
30	18 cm	Measurement	P3	Finds perimeter of a rectangle.
31	6 cm	Measurement	P4	Finds a missing rectangle dimension from area.
32	C or cylinder	Geometry and Shapes	P2	Identifies a 3D shape.
33	4	Geometry and Shapes	P4	Knows a rectangle's angle property.
34	A or parallel	Geometry and Shapes	P3	Recognises parallel lines.
35	Any correct line of symmetry on the square	Geometry and Shapes	P4	Identifies line symmetry. Accept vertical, horizontal, or either diagonal.
36	4	Data and Graphs	P2	Reads a picture graph with scale.
37	Cara	Data and Graphs	P2	Interprets which category has the greatest value.
38	Tue	Data and Graphs	P4	Reads and compares values in

39	32	Word Problems and Mathematical Reasoning	P2/P3	a table. Solves a 2-step addition/subtraction problem.
40	15	Word Problems and Mathematical Reasoning	P4	Finds a fraction of a set.

C. Scoring Sheet

1 point per question. Total: 40 points.

Skill Category	Number of Questions	Question Numbers	Student Score	Notes
Number Sense and Place Value	6	1-6	___ / 6	Look for counting, order, comparison, place value, rounding.
Addition and Subtraction	6	7-12	___ / 6	Check mental strategies and written accuracy.
Multiplication and Division Readiness	5	13-17	___ / 5	Check equal groups, facts, sharing, remainder.
Fractions	5	18-22	___ / 5	Check part-whole ideas, comparison, equivalence, mixed number understanding.
Money and Time	5	23-27	___ / 5	Check money notation, change, clock and duration.
Measurement	4	28-31	___ / 4	Check units, conversion, perimeter, area.
Geometry and Shapes	4	32-35	___ / 4	Check 3D shapes, line properties, symmetry.
Data and Graphs	3	36-38	___ / 3	Check graph/table reading.
Word Problems and Mathematical Reasoning	2	39-40	___ / 2	Check whether student can apply ideas in context.
Total	40	1-40	___ / 40	Use total score together with category profile.

D. Overall Score Interpretation

Overall Score	Placement Meaning	Teaching Interpretation
0-10 points	Beginning / Emerging	Likely needs early number sense, counting, shape recognition, and simple addition support.
11-20 points	Lower Primary Developing	Likely working around P1-P2 level. Needs support with place value, addition/subtraction, time, money, and early multiplication.
21-30 points	Middle Primary Developing	Likely working around P2-P3 level. Can handle basic operations but may need support with fractions, word problems, measurement, and multiplication/division.
31-36 points	Upper Primary Ready	Likely working around P3-P4 level. Ready for more complex word problems, fractions, multi-step operations, and geometry.
37-40 points	Advanced / Needs Enrichment	Likely ready for P4-P5 style extension tasks. Should receive challenge problems involving multi-step reasoning, fractions, area/perimeter, and data.

E. Skill Category Interpretation

Number Sense and Place Value

- 0-40%: Student needs support with counting, comparing numbers, number order, and basic place value.
- 41-70%: Student shows partial place value understanding but may make mistakes with larger numbers or rounding.
- 71-100%: Student is ready for larger numbers, stronger place value work, and more advanced operations.

Addition and Subtraction

- 0-40%: Student needs support with basic facts, number bonds, and simple addition and subtraction.
- 41-70%: Student can solve some number sentences but may make regrouping or place value errors.
- 71-100%: Student is secure with core addition and subtraction skills and is ready for more complex calculation.

Multiplication and Division Readiness

- 0-40%: Student needs concrete practice with equal groups, repeated addition, sharing, and basic facts.
- 41-70%: Student understands the ideas but may be slow or inaccurate with facts

and division.

- 71-100%: Student is ready for stronger multiplication/division fluency and harder problem solving.

Fractions

- 0-40%: Student needs support with fractions as equal parts of a whole and simple fraction language.
- 41-70%: Student understands some fraction ideas but may confuse comparison, equivalence, or simplification.
- 71-100%: Student is ready for broader fraction work, including mixed numbers and fraction operations.

Money and Time

- 0-40%: Student needs support with counting money, reading clocks, and understanding hour and minute ideas.
- 41-70%: Student can manage familiar money and time tasks but may struggle with change, duration, or 24-hour time.
- 71-100%: Student is secure with practical money and time skills and is ready for more complex applications.

Measurement

- 0-40%: Student needs support choosing units, comparing measures, and understanding length or area language.
- 41-70%: Student shows some measurement understanding but may make errors with conversion, perimeter, or area.
- 71-100%: Student is ready for more advanced measurement tasks and multi-step area/perimeter work.

Geometry and Shapes

- 0-40%: Student needs support with naming shapes, recognising properties, and basic spatial language.
- 41-70%: Student recognises common shapes and lines but may be unsure about symmetry or shape properties.
- 71-100%: Student is ready for more advanced geometry, including angles, symmetry, and shape analysis.

Data and Graphs

- 0-40%: Student needs support reading simple graphs or tables and matching data to pictures.
- 41-70%: Student can read some data displays but may miss scale, totals, or comparison details.
- 71-100%: Student is ready to interpret data displays more confidently and explain results.

Word Problems and Mathematical Reasoning

- 0-40%: Student needs support choosing operations, understanding key information, and explaining thinking.
- 41-70%: Student can solve some routine word problems but may struggle with multi-step reasoning.
- 71-100%: Student is ready for more complex contextual problems and can apply skills flexibly.

F. Recommended Grouping Plan

Group	Suggested Profile	Teaching Focus
Group 1: Foundations	Students with very low overall scores or weak Number Sense and Addition/Subtraction results	Counting, number order, comparing quantities, number bonds, simple addition/subtraction, basic shapes.
Group 2: Lower Primary Skills	Students who know basic numbers but still show gaps in place value, money, time, and early multiplication	Place value, addition/subtraction fluency, money, time, simple multiplication and division, graph reading.
Group 3: Middle Primary Skills	Students who are mostly secure with basic operations and are ready for fractions, measurement, and more structured problem solving	Multiplication/division fluency, fractions, perimeter/area, time duration, money, word problems.
Group 4: Extension Group	Students with high total scores and strong category scores across most strands	Multi-step reasoning, fraction of a set, advanced area/perimeter, richer data tasks, and challenge problems.

Quick placement rule

- Place in Group 1 if total score is 0-10, or if Number Sense and Addition/Subtraction are both below 50%.
- Place in Group 2 if total score is 11-20, or if basics are secure but Money and Time, Multiplication/Division, or Measurement are weak.
- Place in Group 3 if total score is 21-30, especially when the student is ready for fractions and longer word problems.
- Place in Group 4 if total score is 31-40 and the student also scores at least 70% in most categories.

G. Short Teacher Notes on How to Use the Pre-Test

1. Give the test in one sitting if possible. About 40-50 minutes is suitable for most students. Younger students may need a little more time.
2. Let students attempt the paper independently first. For very young or hesitant students, you may read questions aloud without helping with the mathematics.
3. Mark by category, not just by total score. A student with a moderate total may still have a major gap in one strand.
4. Use the category profile to plan instruction groups. Re-teach the weakest strands first, especially Number Sense, Addition/Subtraction, and Multiplication/Division readiness.
5. Keep students' written work. Error patterns are useful for follow-up interviews or short oral checks.
6. Reuse a small set of similar questions after intervention to measure growth.